

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

November 2012

EC405 Digital System Design

Date: 8.12.2012, Saturday

Time: 10:00 p.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) What is the use of sensitivity list in VHDL? 02
- (b) Describe the concept of scan path testing. 03
- (c) Explain steps for sequential design process. 02
- Q - 2 (a) List the various advance features of CPLD compared to SPLD. 04
- (b) Design sequence detector for "1011" sequence using mealy state machine in VHDL. 06
- (c) Explain self test circuit for RAM using signature register. 04

OR

- Q - 2 (a) Explain Antifuse and SRAM programmable switching technology in detail. 06
- (b) Explain the Architecture of FPGA with necessary diagram. 04
- (c) Define the following terms: BIST, MISR, s-a-0, s-a-1 04
- Q - 3 (a) Explain with flow chart the complete flow of Digital Design and various approaches of designing. 06
- (b) (i) Draw a schematic diagram for an LFSR with $n=4$ that generates maximum length sequence. 08
- (ii) Add logic so that 0000 is included in the state sequence.
- (iii) Determine the actual state sequence.

OR

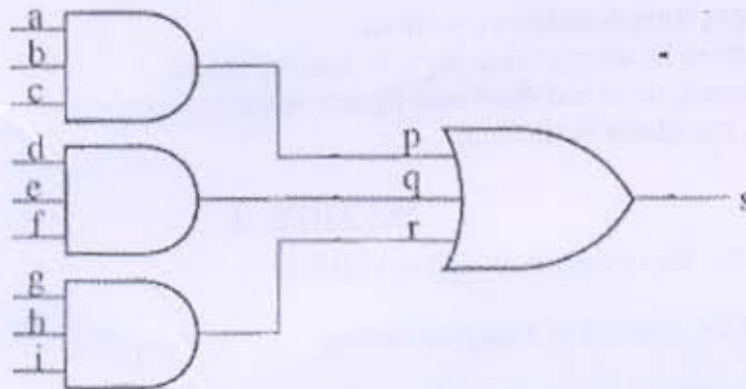
- Q - 3 (a) Explain a 4-bit BILBO register. How the BILBO registers are used for testing a network with two combinational blocks? 08
- (b) Explain boundary scan architecture. 06

SECTION – II

- Q - 4 (a) What are the differences between Melay and Moore sequential machine. 02

- (b) What is GAL? 01
- (c) Discuss process and wait statements. 02
- (d) Differentiate between bit and std_logic in VHDL. 02

Q - 5 (a) Find a minimum set of test vectors that will test all stuck at 1 fault in the following networks. 06

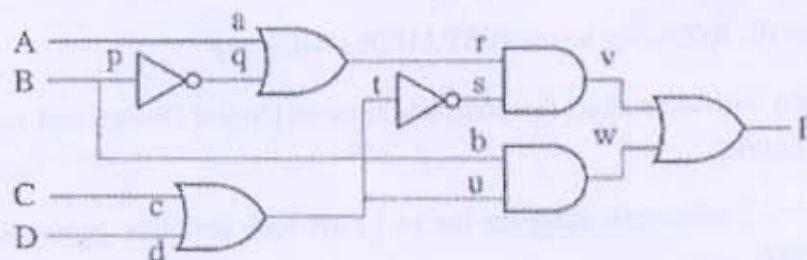


- (b) Explain in brief System on a Chip and System in a Package 04
- (c) Differentiate between Programmable logic and Discrete logic. 04

OR

Q - 5 (a) Design an 8 bit ALU which includes the following features: ADD, SUB, AND, OR, SAL, ROL 08

- (b) Determine the necessary input to the following network to test for u stuck-at-0. For this set of inputs, determine which other stuck at faults can be tested. 06



Q - 6 (a) Write VHDL program for traffic light controller which includes the complete digital circuit explanation. 08

- (b) Explain Antifuse and SRAM programmable switching technology in detail. 06

OR

Q - 6 (a) Explain Verification and validation of an RTL design 06

- (b) Explain the main modules of keypad scanner. 06

- (c) In VHDL Behavioral modeling style is sequential or concurrent? Justify your answer. 02